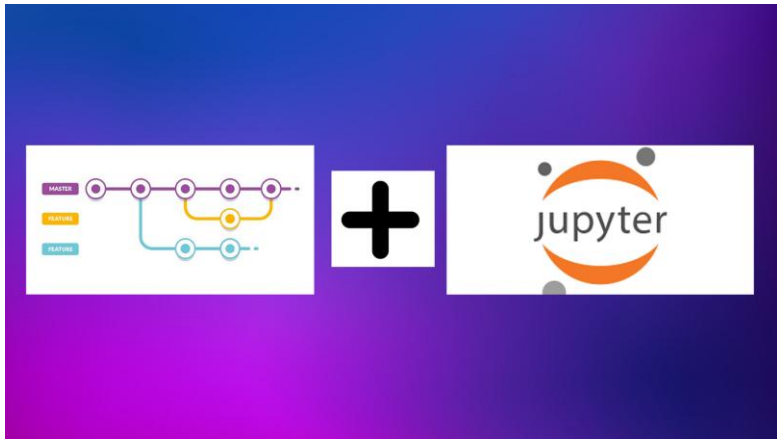


BEYOND THE BASIC SCRIPT



- **The Ecosystem:** Moving from standalone Python scripts to integrated environments like **Jupyter Lab**, **VS Code Notebooks**, and **Google Colab**.
- **Why it matters:** These tools allow for "Literate Programming"—mixing live code, equations, and rich text in one place.
- **Cloud vs. Local:** Using Colab for free GPU/TPU access versus VS Code for heavy local development.

THE VERSION CONTROL CRISIS (NBDIME)



- **The Problem:** Notebooks are stored as complex JSON files. Standard Git "diffs" are unreadable, showing metadata changes instead of code changes.
- **The Solution:** **nbdime** (Notebook Diffing and Merging).
- **Capability:** It allows you to see exactly which code cell or image changed without the "JSON noise".

BRIDGING THE "TECHNICAL GAP" WITH MARKDOWN

- **Stakeholder Communication:** Non-technical managers don't want to read code; they want to read insights.
- **Markdown's Role:** Using headers, bold text, and LaTeX to turn a technical notebook into a professional business report.
- **Executive Summaries:** Every professional notebook should start with a Markdown cell explaining the "Why" and the "Result".

LinearRegression Example

To go over this model, we'll use the `medical_insurance.csv` dataset

- contains information about 1338 people (age, sex, bmi, whether they have children or not, etc.)
- we want to see if we can predict their insurance cost

```
In [165]: 1 import pandas as pd
          2 import numpy as np
          3 from sklearn.preprocessing import OneHotEncoder
          4 from sklearn.preprocessing import StandardScaler
          5 from sklearn.model_selection import train_test_split
          6 from sklearn.model_selection import cross_val_score
          7 from sklearn.linear_model import LinearRegression
          8 from sklearn.metrics import mean_squared_error as mse
          9 from pandas.plotting import scatter_matrix
         10 import matplotlib.pyplot as plt
         11 import seaborn as sn
         12
         13 %matplotlib inline
         14 sn.set()
```

Let's read the data and do some exploratory data analysis

```
In [166]: 1 data = pd.read_csv('data/medical_insurance.csv')
          2 data.head()
```

Out[166]:

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230

KERNEL MANAGEMENT & REPRODUCIBILITY

- **The "Kernel" Concept:** The engine that executes your code.
- **Version Control:** Understanding how to switch between Python 3.8 and Python 3.11 inside the same interface to avoid library conflicts.
- **Environment Isolation:** Ensuring that your "Agricultural" project doesn't break your "Healthcare" project by using separate kernels.

